

Problem-3

Corrected codes

Bug1

```
import requests
```

```
API_BASE_URL = "https://hr-  
internal.company.com/api"
```

```
API_TOKEN = "your_token_here"
```

```
def get_employee(employee_id):
```

```
    url = f"{API_BASE_URL}/employees/{employee_id}"
```

```
    headers = {
```

```
        "Authorization": f"Bearer {API_TOKEN}"
```

```
    }
```

```
    response = requests.get(url, headers=headers)
```

```
    if response.status_code == 404:
```

```
        return None
```

```
data = response.json()
return data.get("employee")
```

Bug2

```
def get_monthly_attendance(conn, employee_id,
month, year):
    query = """
        SELECT a.date, a.check_in, a.check_out, e.name
        FROM attendance a
        INNER JOIN employees e
        ON a.employee_id = e.id
        WHERE a.employee_id = ?
        AND a.month = ?
        AND a.year = ?
        ORDER BY a.date ASC
    """

    cursor = conn.cursor()
    cursor.execute(query, (employee_id, month, year))
```

```
return cursor.fetchall()
```

Bug3

```
from datetime import datetime
```

```
WORK_START = "09:00"
```

```
LATE_THRESHOLD_MINUTES = 15
```

```
def calculate_hours(check_in: str, check_out: str):
```

```
    fmt = "%H:%M"
```

```
    ci = datetime.strptime(check_in, fmt)
```

```
    co = datetime.strptime(check_out, fmt)
```

```
    # Fix: subtract check-in from check-out to get correct  
    duration
```

```
    duration = co - ci
```

```
    hours_worked = duration.seconds / 3600
```

```
    start = datetime.strptime(WORK_START, fmt)
```

```
late_by = (ci - start).seconds // 60
```

```
# Fix: employees exactly 15 minutes late should also  
be flagged
```

```
is_late = late_by >= LATE_THRESHOLD_MINUTES
```

```
return round(hours_worked, 2), is_late
```

Bug4

```
def generate_summary(records):
```

```
    total_hours = 0
```

```
    late_days = 0
```

```
# Fix: iterate only up to the last valid index
```

```
for i in range(len(records)):
```

```
    record = records[i]
```

```
    total_hours += record["hours"]
```

```
    if record["is_late"]:
```

```
        late_days += 1
```

```
# Fix: avoid division by zero when records list is
empty

avg_hours = total_hours / len(records) if len(records)
> 0 else 0

return {
    "total_hours": round(total_hours, 2),
    "avg_hours": round(avg_hours, 2),
    "late_days": late_days,
    "days_present": len(records)
}
```

Bug5

MIN_DAYS_REQUIRED = 20

MAX_LATE_DAYS = 3

```
def check_attendance_policy(summary):
    days_present = summary["days_present"]
    late_days = summary["late_days"]
```

Fix: flag employees who attended fewer than the minimum required days

```
below_minimum = days_present <
MIN_DAYS_REQUIRED
```

Fix: employees with exactly MAX_LATE_DAYS late check-ins should also be flagged

```
exceeded_late = late_days >= MAX_LATE_DAYS
```

```
if below_minimum or exceeded_late:
```

```
    return {
```

```
        "warning": True,
```

```
        "reason": []
```

```
        + (["Below minimum attendance"] if
```

```
below_minimum else [])
```

```
        + (["Exceeded late check-ins"] if exceeded_late
else [])
```

```
    }
```

```
return {"warning": False, "reason": []}
```